Unit 1: Computer System

Introduction of Computer:

Concept of Computer

The development of computer is not a one day's invention. In fact, it began with the civilization and computing instruction on devices. The word **"computer"** comes from Latin word **"computare"**, which means **to calculate**. The word "calculate" mean to find out the result of some operation on numeric and non-numeric values. So, a computer is normally considered to be a calculating device at high speed.

Definition

A computer is an electronic device which accepts raw data and instruction as an input, perform calculations and give result (information). It can even store the output for future use.

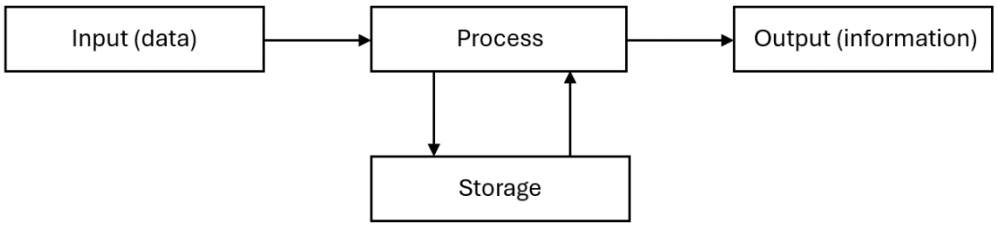


Figure: IPO Cycle

Basic Functions of Computer System

Input: It is a set of data and instructions given to the computer for processing.

Process: It is the conversion of input into output which is done by processing unit (CPU).

Output: It is the result given by the computer after processing the input.

Storage: It is the location where data are stored either temporarily or permanently.

Characteristics of Computer:

1. **Speed:** Computer is a high-speed electronic device capable of processing millions of instructions per second (MIPS), even billions of instructions per second (BIPS). The speed of computer is measured in terms of Gigahertz and Megahertz.

Measuring Units in Frequency	Measuring Units in Time
1 Hz (Hertz or 1×10^0 Hz)	1 Second (1×10^0 sec)
1 KHz (Kilo Hertz or 1×10^3 Hz)	1 milli second (1×10^{-3} sec)
1 MHz (Mega Hertz or 1×10^6 Hz)	1 micro second (1×10^{-6} sec)
1 GHz (Giga Hertz or 1×10^9 Hz)	1 nano second (1×10^{-9} sec)

1 THz (Tera Hertz or 1×10^{12} Hz)	1 pico second (1×10^{-12} sec)
1 PHz (Peta Hertz or 1×10^{15} Hz)	1 femto second (1×10^{-15} sec)
1 EHz (Exa Hertz or 1×10^{18} Hz)	1 atto second (1×10^{-18} sec)
1 ZHz (Zeta Hertz or 1×10^{21} Hz)	

2. **Accuracy:** Computer perform calculations with 100% accuracy. Computer never make any mistakes if the given input is correct and the program and hardware devices are correct. But sometimes we get wrong output due to the wrong input and errors in program which is known as GIGO (Garbage In Garbage Out).

3. **Versatility:** Computer is versatile in nature. It can perform various tasks depending upon the different program. It has a wide range of application areas. Some of the application areas of computers are in education, bank, office, medical diagnosis, science and technology, communication and astronomy etc.

4. **Diligence:** Computer is capable of performing same task continuously without feeling bore and tired with same speed and accuracy. That's why it is known as diligent machine.

5. **Automatic:** Computer is an automatic machine, capable of functioning automatically, once the appropriate set of instructions (program) and data are provided to the computer.

6. **Reliability:** Computer is reliable because it gives 100% accurate result if the input is correct. It can work for a long period of time without getting tired. That's why, they are widely used everywhere.

7. **Storage Capacity:** A computer can store large amount of data. It has a storage area, called memory. Date stored in the memory can be retrieved at any time and at very fast speed. The memory can be divided into primary memory and secondary memory. Example: RAM, ROM, floppy disk, hard disk, magnetic tape etc. The storage capacity of computer is measured in terms of byte, kilobyte, megabyte, gigabyte etc.

Unit	**In Unit**
1 Bit	0 or 1
1 Byte	8 bits
1 Nibble	4 bits
1 Kilo byte (KB)	1024 bytes
1 Mega byte (MB)	1024 Kilo bytes (KB)
1 Giga byte (GB)	1024 Mega bytes (MB)
1 Tera byte (TB)	1024 Giga bytes (GB)

Application of Computer:

Today computer has become an essential part of our life. It can be found everywhere in office, schools, homes, hospitals, private and government companies and many more places. Some of the application areas of computers are: -

1. **Computer in Education:** Computer are used in education for teaching and learning. It is used for providing Audio-Visual learning, research, online learning, maintain records, make documents etc. Now a days there are thousands of websites available over internet to deliver education for free in almost every subject.

2. **Computer in Business:** In business sector computer are used for keeping records, analyzing sales, budgeting, payroll management, marketing products, performing transactions. Computer are mostly used in the department malls, shops, hotels etc. E-commerce is very popular today as it involves buying and selling goods or services via internet.

3. **Computer in Office:** In office computer is used in preparing report, letters, contract, form, note and notice that is the basic works of an office. It helps user in preparing, storing, retrieving and displaying text. Computer is used in the most of the departments in office. For example: administration, account, marketing and production.

4. **Computer in Bank:** Computer is used in bank for general computations, handling transaction, to maintain ledger, to issue fixed deposit receipts, to make withdrawals and deposits to customers. The large volumes of cheque received everyday in banks are processed by computer. These days bank provide 24 hours ATM service, where user can withdraw their money anytime from ATM.

5. **Computer in Health and Medical Field:** Computer are widely used in hospitals to help doctors in diagnosis, getting information on patient's diseases, required treatments, keeping patient's record. To conduct various surgeries junior doctors, get help from another professional doctor by web conferencing. Different kinds of medical equipment such as Ultrasound machines, CT scan, MRI scan are controlled by computer.

6. **Computer in Airlines:** In airlines, computer stores all necessary information regarding ticket reservation, such as flight number, flight route, flight distance, number of seats available for each flight in each class. It also stores information regarding passengers. It also gives information regarding flight departure and arrival time.

7. **Computer in Entertainment:** Computer are widely used for an entertainment purpose. We can play various computer games, watch movies, TV shows, listen to music. We can also download and share them on the internet.

8. **Computer in Communication:** A computer is widely used in communication. Communication is done by email, social media, online conferencing with the help of internet. It has made the communication faster, easier and cheaper.

9. **Computer in Military Defense:** Computer is widely used in defense purposes. At present, sophisticated weapons such as fighter jet, missiles, tanks are being widely used in military operations. Computer are used to track fighter jet, missiles, tanks using radar system and destroy them by air defense system before it comes on the surface.

10. **Computer in Graphics:** Computer is being used to draw drawings, pictures, design logo and banner. Different graphic software packages like Adobe Photoshop, Adobe Illustrator, Corel Draw are used to draw sophisticated drawings. We can also edit pictures and modify them according to our requirements.

Evolution of Computer Technology:

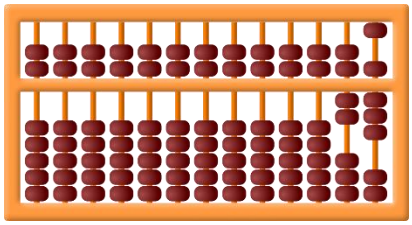
"Necessity is the mother of invention". The saying holds true for computers also because computers were invented as a result of man's search for fast, accurate and comfort calculating devices.

Evolution of computer is a study of past development of computer.

Ancient people lived on earth for centuries without counting. Then they started to count with their ten fingers. It became so difficult to live and to remember them more and more facts using their ten fingers. These phenomena were gradually replaced by the use of stones, sticks, counting notches (V-shaped cut in or on something).

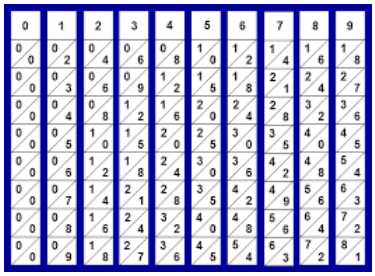
Mechanical Calculating Era

1. **Abacus:** Abacus is a first calculating machine developed by the Chinese in 3000 years ago. It is known as **Suan Pan** in Chinese language and **Soroban** in Japanese. It was used for performing simple calculation like counting, addition, subtraction and multiplication of numbers.



   It consists of a rectangular wooden frame with horizontal rods which carry round beads, Mid-Bar divides each of these rods into upper and lower part. The upper part is called heaven which consists of two beads and each beads equals to value 5. The lower part is called earth which consists of five beads and each beads equal to value 1. Counting is done by shifting the beads from one side to another.

2. **Napier's Bone:** The Scottish mathematician John Napier first published the table of logarithms in 1614 A.D. Then based on his own principle, he invented Napier bone in 1617 A.D. It is used to perform division by subtraction and multiplication by adding, according to the principle of logarithms.



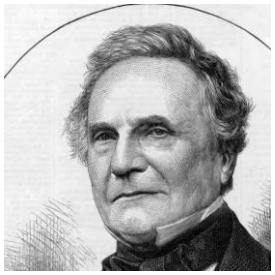
3. **Slide Rule:** The English man William Oughtred invented a rectangular device slide rule in 1620 A.D. It was a calculating device based on the principles of logarithm. It consists of two set of rulers sliding by one another to perform multiplication and division.

4. **Pascaline:** The French mathematician Blaise Pascal invented the Pascaline, the first mechanical calculating machine capable of performing addition and subtraction. It used gears and wheels to calculate. Its purpose was to assist his father in his accounting work (tax calculation).

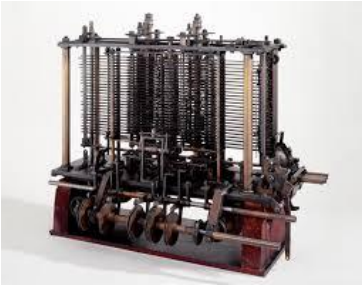
5. **Stepped Reckoner:** The German mathematician and philosopher Baron Gottfried Wilhelm Von Leibniz modified the Pascaline machine and invented Stepped Reckoner. It was able to perform automatic addition, subtraction, multiplication, division and could find out square root. it used fluted drums instead of gears. This machine was named as Leibnitz's calculator.

6. **Jacquard's Loom:** The French man Joseph Marie Jacquard was a textile manufacturer who invented a mechanism for automated weaving cloths for the textile industry at Lyon in 1802 A.D. This machine was used to automatically control weaving looms to facilitate the production of weaving cloth with complex patterns. This machine was controlled by punch cards i.e. principle of present and absence of holes.

7. **Difference Engine and Analytical Engine:** The English professor and mathematician Charles Babbage invented the **Difference Engine** in 1822 A.D. It was designed to solve different mathematical equations and calculate various mathematical function, logarithmic tables, polynomial and trigonometric functions. The project could not be completed due to lack of funds. Babbage continue working on his difference Engine for a full ten years but he had a better idea.



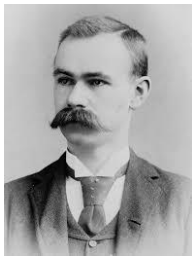
His new idea was the construction of a general purpose, fully programmable automatic mechanical counting machine. Babbage called his machine an **Analytical Engine** in 1833 A.D. The idea of Analytical Engine was considered as a most significant step in the history of computers, because his idea only, the dream of modern computer has come true. This machine had five units: input, output, store, mill and control which are similar to the units of a modern computer. That's why he was considered as the "**Father of computer**". Unfortunately, due to lack of technology, Analytical Engine was never completed.



8. **Lady Augusta Ada Byron Lovelace:** The English intelligent and independent-minded woman – Lady Augusta Ada Byron Lovelace, was daughter of English poet Lord Byron and a very great follower, assistant of Charles Babbage. She suggested Charles Babbage to use binary number system in Analytical Engine. She gave the concept of computer programming for the first time, so she is considered as **First computer programmer**. The US Department of Defense developed programming language named Ada in her honor in 1979 A.D.

9. **George Boole:** George Boole was a self-taught English mathematician, logician and Philosopher. He developed an algebra based on two values (true and false) which is known as Boolean algebra. Since computer work in binary, two values (1 and 0), Boolean algebra became the natural way to design the logic of computer hardware.

10. **Tabulating Machine:** An American inventor Herman Hollerith invented a machine known as **Tabulating Machine** in 1886 A.D. It used the method of card punching to storing the data, which is fed into a machine that compiled the result mechanically. The machine was designed to compute the US census record of 1890 A.D. It was able to complete compilation in 3 years which earlier used to take around 10 years. He formed the Tabulating Machine Company in 1896 A.D. to manufacture his inventions. The Tabulating Machine Company and other two companies merged in 1924 A.D to form International Business (IBM) which is the largest computer manufacturing company in the world.



The Electro Mechanical Era

1. **Mark – I:** The professor Howard Aiken along with his staff invented Mark-I in 1937 A.D. It was a first electro mechanical computer. It is also known as ASCC (Automatic Sequence Controlled Calculator). It was 51 ft. long, 8 ft. tall and weighted nearly 5 tons. This machine took 1 second to perform 3 mathematical calculations. It was used in military purposes, including development of the atomic bomb.

2. **Atanasoff Berry Computer (ABC):** In 1938, John J, Atanasoff and Clifford Berry designed ABC for solving systems of simultaneous equation. Is used 18000 Valves and other 5 valves for internal logic and capacitors for storage of electrical charges. It used punched cards as secondary storage.

3. **John Von Neumann:** The Hungarian mathematician, John Von Neumann gave the idea of stored program computer in the sense that program is stored internally in the main memory of the computer along with its associated data in 1945. So, he is called the **Father of Stored program**. Before that, program required for the computer were integrated and written permanently in chips, so modification of program was not possible. But after Neumann, such programs were stored inside computer in some storage media, so modification was easy and flexible.

The Electronic Era

1. **ENIAC (Electronic Numerical Integrator and Calculator):** John William Mauchly and John Presper Eckert invented ENIAC in 1946 A.D. It was the first general purposed electronic computer. It contained 18000 vacuum tubes, 15000 relays and thousands of resistors and capacitors. It weighted about 30 ton consumed 200 kilowatts of electric power. It was based on decimal number system. It could add two numbers in 200 microsecond and multiplication of 10-digit numbers in 3 millisecond.

2. **EDSAC:** EDSAC (Electronic delay storage automatic computer) was invented by professor Maurice Wilkes in 1949 A.D. It also used vacuum tubes. Although started after EDVAC, it was completed before EDVAC. So, it became the first stored program computer.

3. **EDVAC:** EDVAC (Electronic Discrete Variable Automatic Computer) was developed by John Presper Eckert and John William Mauchly in 1952 A.D. and used for more school personnel and the Ballistics Research Laboratory of the USA Army. EDAVC was based on John Von Neumann's ideas. It also used vacuum tube and some internal storage.

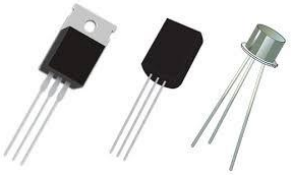
4. **UNIVAC:** UNIVAC was developed by John Presper Eckert and John William Mauchly in 1961 A.D. It was the first computer manufactured for commercial use and general-purpose digital computer. Before this all the computers were either used for defense or census.

Generation of Computer:

The average time period in which old computer technology is replaced by newer technology is known as generation of computer. There are 5 generation of computer which are as follows:

1. **First Generation Computer:** In first generation, vacuum tubes were used as CPU components and magnetic drum for storage and machine language was used for giving instructions. The computer of this generation was very large in size. The power consumption was very high and generated much heat. The programming of first-generation computer was done in machine language. The UNIVAC and ENIAC computers are examples of first-generation computing devices.

2. **Second Generation Computer:** In second generation, transistor replaced vacuum tubes. Transistors were highly reliable compared to vacuum tube. The size of computer was reduced as compared to first generation and less power consumption and heat generation. The speed was much higher than first generation computer which was measured in micro second. Magnetic core was used as primary and magnetic drum as secondary memory. Assembly language and machine independent language (high level) such as COBOL and FORTAN were introduced.



3. **Third Generation Computer:** In third generation, transistors were replaced by integrated circuit (IC). Scientists manage to fit many components on a single chip; as a result, computers became ever smaller as more components were squeezed on the chip. Magnetic disks began to replace magnetic tape for secondary and video display terminals were introduced for the output of data. Keyboard and monitors were introduced for data and output respectively. They had high storage capacity.

4. **Fourth Generation Computer:** The invention of microprocessor chip marked the beginning of the fourth-generation computers. Semiconductor memories replaced magnetic core memories. The development of chips decreased the size and price of computers. The invention of microprocessors led to the development of microcomputer or the personal computer. The first microprocessor called Intel 4004 was developed by American Intel Corporation in 1971.

This computer has faster accessing and processing speed and increased memory capacity. These features led to the development of much more powerful operating system. Developments in fourth generation language and application software for microcomputers became popular and allowed home and business user to adopt their computers for word processing, spreadsheet manipulating, file handling and graphics.

5. **Fifth Generation Computer:** Fifth-generation computers are still in development, are built on artificial intelligence and advanced microelectronics. The fifth generation of computers is designed to be much faster, more powerful, and more intelligent than any other generation. The fifth generation of computers understands the natural human language. Researchers are planning to produce biochips that are to be used in fifth generation computer. While a truly fifth generation computer is not available, the computer scientists are implementing fifth generation concept on fourth generation computer.

Table: Generation of Computers with their main features.

	First Generation	Second Generation	Third Generation	Fourth Generation	Fifth Generation
Electronic Device	Valve	Transistor	IC	Microprocessor	Bio-Chip
Speed	Mili Second	Micro Second	Nano Second	Pico Second	More than Femto Sec
Internal Memory	Valve	Magnetic Core Memory	Semi Conductor Memory	Semi Conductor Memory	Super Conductor Memory
Permanent Storage	Punch Card	Magnetic Tape	Magnetic Disk (Floppy Disk)	Magnetic and Optical Disks (Hard Disk, CD, DVD, Pen Drive)	New technology for storage
Computer Language	Fully Machine Level	Assembly language and little bit of High level language	Fully High Level	High Level and 4GL	Natural Language
Cost, Size, Electrical consumption, Heat generation	Very high	High	Low	Very Low	Not Known
Speed, Capacity	Very less	Less	Moderate	High	Much Higher

Measurement Unit of Processing Speed and Storage Unit:

Unit of Processing Speed: Computer speed refers to the CPU speed. A CPU is the device that interprets and executes instruction. The CPU has to fetch, decode and execute instruction to transfer information and receive from other resources. There are different measurement unit for computer processing speed. CPU speed is usually measured in the unit of hertz, MIPS and FLOPS.

The measurement unit of microcomputer is Hertz (Hz). Hertz is the unit of frequency measured cycles per second. One hertz is one complete cycle in one second. Similarly, 3 GHz is 3 x 1000 x 1000 x 1000 cycles in one second.

The different measurement units and their relationship for microcomputer speed are as follows:

Measuring Units in Frequency	**Measuring Units in Time**
1 Hz (Hertz or 1×10^0 Hz)	1 Second (1×10^0 sec)
1000 Hz = 1 KHz (Kilo Hertz or 1×10^3 Hz)	1 milli second (1×10^{-3} sec)
1000 KHz = 1 MHz (Mega Hertz or 1×10^6 Hz)	1 micro second (1×10^{-6} sec)
1000 MHz = 1 GHz (Giga Hertz or 1×10^9 Hz)	1 nano second (1×10^{-9} sec)
1000 GHz = 1 THz (Tera Hertz or 1×10^{12} Hz)	1 pico second (1×10^{-12} sec)
1000 THz= 1 PHz (Peta Hertz or 1×10^{15} Hz)	1 femto second (1×10^{-15} sec)

The measurement unit for mainframe and mini computer is in MIPS(Millions Instructions Per Second) rather than hertz.

Modern mainframe computers execute millions of instructions in one second. The processing speed of these computers is directly measured on how much they execute instruction in a single unit of time rather than cycles per second

The processing speed of super computer is measured in “FLOPS” (Floating Point Operation Per Second).

Storage Unit: Computer store information in the form of “1” and “0” in different types of storage such as memory, hard disk, USB drives etc.

The most common digital data storage unit is byte which is 8 bits.

Computer data is expressed as bytes, kilobytes, megabytes as it is in the metric system, but 1 kilobyte is 1024 bytes not 1000 bytes.

Data storage units are: bit(b), byte(B), kilobyte (KB), megabyte (MB), gigabyte (GB), terabyte (TB), petabyte (PT), Exabyte (EB), zettabyte (ZB), yottabyte (YB).

1. **Bit:** A bit (binary digit) is the smallest unit of data that a computer can process and store. A bit is always in one of two physical states, similar to an on/off light switch. The state is represented by a single binary value, usually a 0 or 1. However, the state might also be represented by yes/no, on/off or true/false. Bits are stored in memory through the use of capacitors that hold electrical charges.

2. **Nibble:** A NIBBLE is a string of four BITS. i.e. 1 Nibble is equal to 4 bits.

3. **Bytes:** It is the unit of measuring computer’s memory and size of a program. One Byte is the string of 8 binary digits (BITS). It may represent single letter, numeral or other character.

Unit	In Unit
1 Bit	0 or 1
1 Byte	8 bits
1 Nibble	4 bits
1 Kilo byte (KB)	1024 bytes
1 Mega byte (MB)	1024 Kilo bytes (KB)
1 Giga byte (GB)	1024 Mega bytes (MB)
1 Tera byte (TB)	1024 Giga bytes (GB)
1 Peta byte (PB)	1024 Tera byte (TB)
1 Exa byte (EB)	1024 Peta byte (PB)
1 Zetta byte (ZB)	1024 Exa byte (EB)
1 Yotta byte (YB)	1024 Zetta byte (ZB)

Super, Mainframe, Mini and Micro Computers:

Computer are classified according to *working principle, size, brand,* and *model*. A single computer can be microcomputer on the basis of size, IBM compatible on the basis of brand, AT computer on the basis of model and digital computer on the basis of working principle.

Hence, computer is classified into different types on the basis of following factors:

On the basis of Working Principle	On the basis of Size
• Analog Computer • Digital Computer • Hybrid Computer	• Super Computer • Mainframe Computer • Mini Computer • Micro Computer
On the basis of Brand	**On the basis of Model**
• IBM PC • IBM Compatible Computer • Apple/Macintosh Computer	• XT Computer • AT Computer • PS/2 Computer

On the basis of Size

Super Computer: Super computer is the largest, fastest and most power computer among digital computer. It is a special purposed computer capable of handling huge amount of calculations and complex calculation as well. It can perform at billions of instructions per second (BIPS) and more.

This computer is the most expensive computer (cost about 5 lakhs to 200 million dollars range). It uses parallel processors.

It is used only by large corporations, universities and government agencies, mainly in the areas of defense weaponry, weather forecasting, scientific research, aeronautics, satellite communication and space administration, nuclear research work, petroleum research work, molecular modeling, study of DNA structures.

It requires large space and temperature-controlled room. Example: CRAY X-MP/24, NEC-500, PARAM, ANURAG, Fujitsu VP-2600/10 etc.

Application areas of super computer

- **Biology Areas:** Mostly, supercomputer used to diagnose for various diseases, and provide the assistance for producing good result in strokes, brain injuries and other blood flow issues in your body.
- **Military and Defense Missions:** Supercomputing help to provide virtual testing for nuclear explosion and weapon ballistics.
- **Climate Patterns**: Supercomputer application is able to study and understand climate patterns.
- **Airlines Industry:** With the help of supercomputer, designed the flight simulators for newbie pilots and this simulator help to training for new pilots.
- **Weather Forecasting:** To gather the information related to weather forecasting, supercomputer run in the NOAA's system, means National Oceanic and Atmospheric Administration. NOAA system is able to execute all types of simple and logically instructions.
- **Scientific Research:** In the weather and science research areas depend on the supercomputer because for analyzing data from the exploring solar system, satellites that rounding earth, and other area such as nuclear research.
- **Advance database (Data Mining):** Some large scale companies need the supercomputer for extracting useful information from data storage house or in the cloud system. Such as insurance companies.
- **Simulated Environment in Automobile:** Supercomputer provides the help to people for buying vehicle because before purchasing the vehicle customer can test through simulation environment that is created by supercomputer.
- **Smog Control System:** Scientists use supercomputers in own laboratory for predicting the fog and other pollution level on the particular areas, and then take final step to prevent them.

Mainframe Computer: Mainframe computers are large sized expensive computers capable of supporting hundreds or thousands of users simultaneously. They are specially designed to allow large numbers of users and applications to rapidly and simultaneously access the same data without interfering with each other.

Mainframe computer is used in large organizations such as bank, insurance companies, airlines, railway reservation system etc. Some of the popular companies that use mainframes are Coca Cola, Wal-Mart and Nike.

Examples: IBM1401, IBM7090, ICL2950/10 etc.

Application areas of mainframe computer

- Used to process large amount of data at very high speed such as in the case of Banks/Insurance companies/Hospitals/Railways, which need online processing of large number of transactions and requires massive data storage and processing capabilities.
- Used as controlling nodes in WANs (Wide Area Networks).
- Used to manage large centralized databases.
- It is used in airline reservation and railway reservation system.

Mini Computer: Mini computer is the medium-sized general purposed computer which is smaller than Super and Mainframe computer and larger than Microcomputer. It supports up to 50 terminals. These computers are less expensive than mainframe computer and hence suitable for those organizations that cannot afford high priced server. This computer is useful for small business organizations, industries, universities, banks, engineering etc. It can be used as a server in LANs (Local Area Network).

Example: Prime series, AS/400, AP-3 etc.

Micro Computer: A computer, which is based on microprocessor is called microcomputer. It is also known as Personal Computer. It is a small, low-cost digital computer. It requires small space, can be placed on a table or even kept inside a briefcase. This computer has central processing unit on a single chip. It is mainly use in office, house, school, shop and store.

Example: IBM PC, Apple/Macintosh.

Micro computers can be classified as:

1. Portable Micro computer
    - Laptop and Notebook,
    - Personal Digital Assistant (PDA) and Palmtop,
    - Tablet and Smartphone.

2. Non-Portable Micro computer
   - Desktop/Personal computer,
   - Workstation and Server Computer.

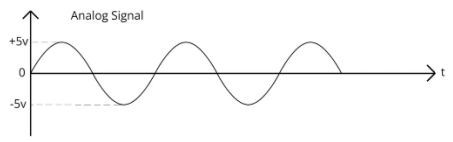
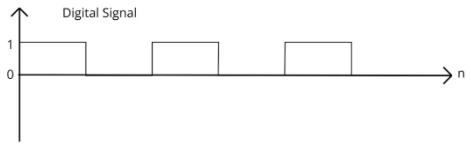
On the basis of Working Principle

Analog Computer: The computer which works on physical values (continuous data) such as voltage, pressure, temperature, length etc. is known as analog computer. It is a special purposed computer and is mainly used in scientific work. Generally, it has less storage capacity and its accuracy is poor compared to digital computer. For example, Presley, thermometer, barometer, speedometer and ammeter.

Digital Computer: The computer which works on discrete data (discontinuous data, binary system 0 or 1) is known as digital computer. It is a general purposed computer. It is faster, more accurate and has larger memory capacity. It can be used in many fields such as bank, hospitals, education, research, entertainment etc.

Hybrid Computer: The computer which has a combined feature of both analog and digital computers, is known as hybrid computer. It can perform the tasks of an analog computer as well as a digital computer. It can convert data from analog to digital and vice versa. It has high cost. It is a special purposed computer. For example: CT-Scan, MRI etc.

Difference between Analog and Digital Computer

Analog Computer	Digital Computer
Analog computer works with continuous values.	Digital computer works with discrete values.
It is slower than digital computer.	It is faster than analog computer.
It is less accurate than digital computer.	It is more accurate than analog computer.
It is special purposed computer.	It is general purposed computer.
It cannot be reprogrammed.	It can be reprogrammed.
No any or smaller storage capacity.	Large storage capacity (memory).
Lower cost compared to digital computer.	Higher cost compared to analog computer.
Output is continuous.	Results are obtained after complete computation.
Wave form of analog computer 	Wave form of digital computer 

Mobile Computing and its Application

Mobile computing is the use of portable devices, such as smartphones and tablets, to access information and perform computing tasks while on the go. The widespread availability of high-speed mobile networks (Wi-Fi or cellular wireless technology, such as 5G.) and advancements in mobile technology have made it possible for people to stay connected to the internet and access digital services from anywhere at any time.

Some of the applications of mobile computing are as follows:

- E-mail
- Contact Management
- Scheduling
- Web Browsing
- Messaging
- Entertainment
- Presentation and many more.

Computer System and I/O Devices

Computer systems is a combination of hardware components (Central Processing Unit (CPU), memory unit, input unit and output unit) and software (programs and applications) that work together to perform specific tasks.

Input Unit: The input unit allows the user to give raw data and instructions to the computer.

Output Unit: Output unit provides final output in human understandable/readable form.

Central Processing Unit: The CPU performs many operations and controls the whole computer system.

Memory Unit: Memory unit are used to hold data/instruction permanently or temporarily as per the requirement.

Concept of Computer Architecture and Organization

The design, arrangement, construction or organization of the different parts of a computer system is known as Computer architecture.

It is a blueprint for the design of a computer system and describes how computer software and hardware interact to make computer function.

We see that a computer system is made of various components and by connecting various peripherals such as mouse, keyboard, speaker, printer, monitor etc.

Hence, the term ‘Computer Architecture’ is the conceptual design and fundamental operational structure of computer system.

Components of Computer System:

The components of computer system are:

Input Unit: Input Unit provides an interface between the users and the computer for giving data and instruction into the computer memory. Some of the common examples of input devices are keyboard, mouse, scanner, light pen, joystick, microphone, and barcode reader etc.

Central Processing Unit (CPU): The CPU is the main processing unit of computer. It is also known as 'brain of a computer'. Its primary function is to execute instruction. It also controls the operations of other units of the computer system.

The different parts of the CPU are:

- Arithmetic and Logic Unit (ALU)
- Control Unit (CU)
- Register Array

Arithmetic and Logic Unit: The arithmetic and logic unit perform arithmetic and logical operations. Arithmetic operations are the fundamental math operations: addition, subtraction, multiplication, and division. Logical operations are comparisons. That is, it compares two pieces of data to see whether one is equal to each other, greater than, or less than.

Control Unit: Control unit controls and coordinates all the hardware of a computer system. It generates the control signal and send it to other parts of a computer to execute the instruction. It acts as a central nerve of the computer system.

Register Array: A register is a small temporary storage unit inside a computer's CPU. It temporarily holds the data required by the CPU for immediate processing.

There are different types of registers are: -

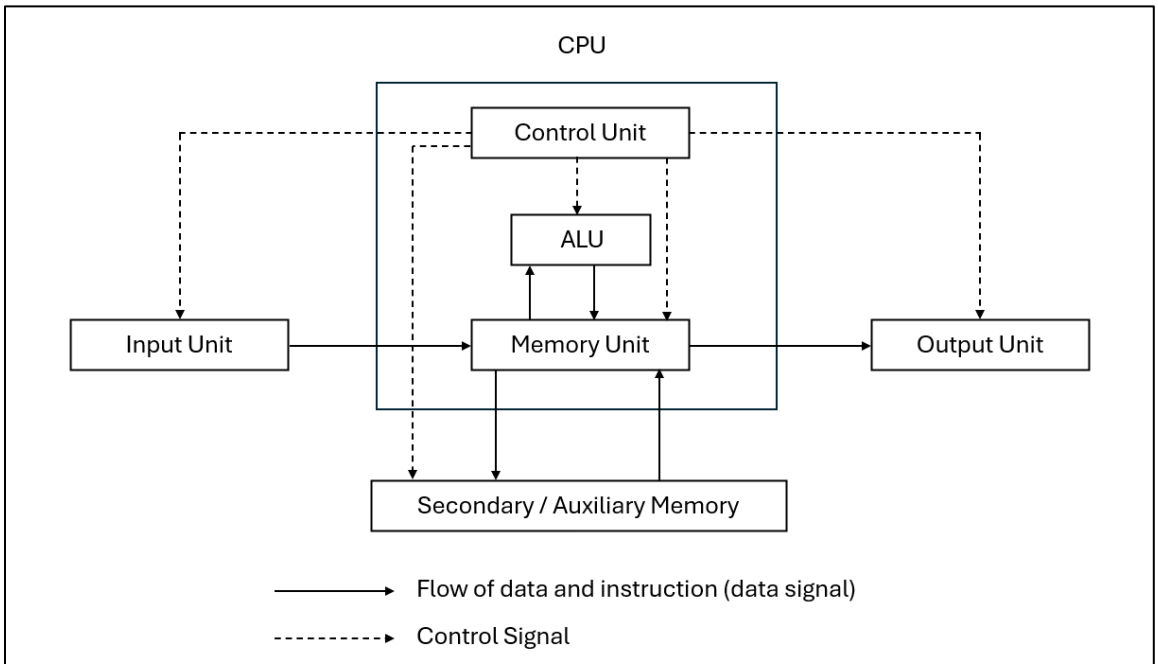
- **Accumulator Register (AR):** It stores the immediate results of arithmetic and logical operations.
- **Program Counter (PC) Register:** It holds the memory address of the next instructions to be executed.

- **General-Purposed Register:** It holds the data and memory address that are actively being processed
- **Instruction Register (IR):** It holds the current instruction being executed.
- **Memory Address Register (MAR):** It holds the memory location of data that needs to be accessed.
- **Memory Data/Buffer Register (MDR):** It holds the data that is being transferred between memory and CPU.
- **Input/Output Register (IO Register):** It is used to communicate with input and output devices.

Output Unit: Output unit consists of devices that translate information processed by the computer into a form that humans can understand. The computer-processed information consists of 0's and 1's, which needs to be translated into words, numbers, or pictures that people can understand. The two types of output devices are Hardcopy output device, Softcopy output device. For example, printer, monitor, speaker etc.

Memory Unit: The memory unit is part of the computer that holds data and instructions for processing. It holds raw data given from the input devices before being processed by CPU. It also stores the result after processing and sent it to output devices. For example, RAM, Hard Disk, Floppy Disk etc. Storage unit is divided into two types:

Primary Storage
Secondary Storage



Block Diagram of Computer System

Bus System

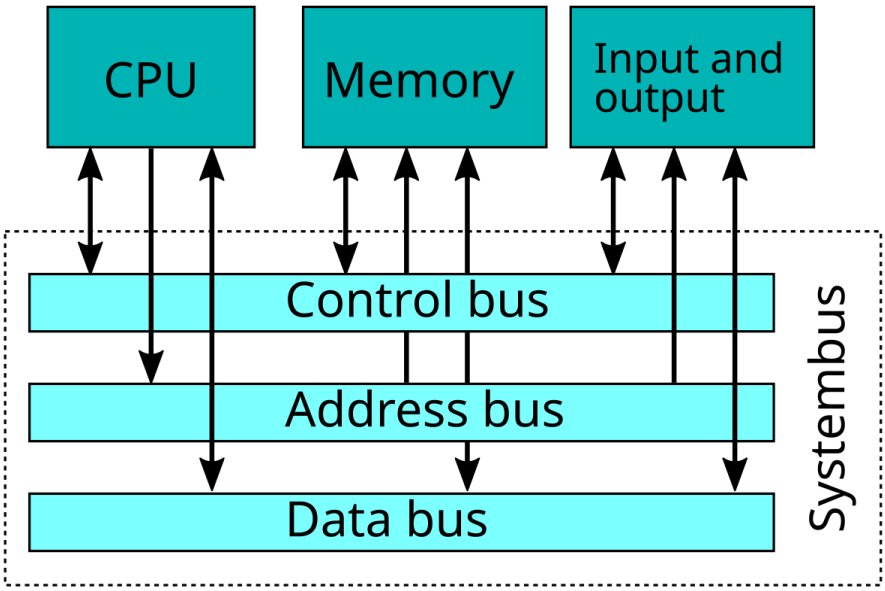
A bus is a collection of electrical pathways through which data, addresses and control signals are transmitted between different hardware components.

Types of Bus:

1. **Data Bus:** Data bus carries data from one component to another. It is uni-directional for input and output devices and bi-directional for memory and CPU.

2. **Control Bus:** Control bus carries control signal, CU of CPU uses control signal for controlling all the components. It is uin-directional from CPU to all other components.

3. **Address Bus:** Address bus carries memory address. A memory address is a numerical value used for identifying a memory location. Computer performs all its task through the memory address. CU of CPU sends memory address to all the components. So, address bus is also uin-directional from CPU to all other components.



Memory

Memory is the part of the computer system that is used to store data or instruction (program) required during the processing of data, and output results.

From the technological point of view, memory can be classified as:

- Semiconductor memory
- Magnetic memory

- Optical memory

Semiconductor memory is faster, lighter and smaller in size and consume less power. It is used as main memory of the computer.

Magnetic and optical memory is slower but cheaper than semiconductor memory.

Memory is divided into two categories:

- Primary Memory
- Secondary Memory

Primary Memory

Primary memory is the main memory or internal memory of computer. It holds data and instructions currently being processed by CPU. It is the memory that is directly associated with the CPU. The semiconductor memories are used as the primary memory. They are: RAM, ROM and cache memory.

RAM (Random Access Memory)

RAM stands for Random Access Memory. It is used to store data that the computer is currently using. It is a read and write memory of the computer. It is a volatile memory i.e. it stores data or information as long as power supply is on, when the power supply goes off, the data stored in the RAM will be lost. It is of two types: -

- **SRAM (Static Random Access Memory)**
  It is a type of RAM made up of flip-flop circuit containing transistors in a memory cell. It stores data or programs as long as power supply is on. It doesn't require refreshing circuit periodically. It is used as cache memory.
- **DRAM (Dynamic Random Access Memory)**
  It is a type of RAM made up of one transistor and one capacitor in each memory cell. It stores data or programs for short period (few millisecond) even though computer is powered on. It needs to be refreshed periodically to retain data or program. It is used as main memory.

Difference between SRAM and DRAM

SRAM	DRAM
SRAM stores data or programs as long as power supply is on.	DRAM losses its information in very short time (few millisecond) even though the power supply is on.
It doesn't require refreshing circuit.	It needs to be refreshed periodically to retain data or program.
It has higher speed compared to DRAM	It has lower speed than SRAM

It stores data in a flip-flop circuit containing transistors in a memory cell.	It requires less number of transistors per memory cell because capacitors and one transistor are needed to form a memory cell.
It stores bit as voltage.	It stores bit as charge.
It is expensive in cost.	It is cheaper in cost.

ROM (Read Only Memory)

ROM stands for Read Only Memory. It can only perform the read operation; it doesn't have write capabilities. It is non-volatile memory, i.e. the information stored in it is permanent and is not lost even if the power supply is cut down. Its content is written at the time of manufacture which cannot be changed but can be read. It is of three types: -

- **PROM (Programmable read only memory)**
  PROM is a programmable ROM. It is once programmable i.e. the user can write data in a PROM only once. User can store permanent programs, data or any kind of information in PROM. Special equipment called ROM programmer or ROM burner is available for storing data in PROM.
- **EPROM (Erasable programmable read only memory)**
  EPROM is an erasable PROM. The data stored in EPROM can be erased by exposing it to ultraviolet rays. When an EPROM is exposed to ultraviolet light, the entire data are erased. It is usually reprogrammed without removing it from the circuit board.
- **EEPROM**
  EEPROM is an electrically erasable PROM. It can be erased and reprogrammed on the byte by byte basis. The program is erased by exposing it to an electrical charge. It requires much shorter time, a few milliseconds for erasing as compared to EPROM.

Difference between RAM and ROM

RAM	ROM
RAM stands for Random Access Memory.	ROM stands for Read Only Memory.
It allows both read and write operations.	It allows only read operation.
It is volatile in nature.	It is non-volatile in nature.
It stores currently running programs of computer system.	It stores firmware and system related program of computer system.
It usually has higher memory space than ROM.	It usually has less memory space than RAM.
Writing data on RAM is faster than ROM.	Writing data on ROM is slower.
The second operation after booting the computer system is performed in RAM.	The first operation in computer system is performed in ROM (during booting process).

RAM can be accessed by the CPU directly	The data of ROM needs to be transferred to RAM to give access to the CPU to its data.
RAM is expensive compared to ROM.	ROM is cheaper in comparison to RAM.
For example: SRAM, DRAM.	For example: PROM, EPROM, EEPROM.

Cache Memory

Cache memory pronounces as cash, is a special, high speed and expensive semiconductor memory placed between RAM and CPU. It is used in CPU, so that CPU doesn't have to wait for data to be delivered. The most frequently used instructions are kept in the cache memory so that the CPU can look there first. This allows the CPU to run faster, because it doesn't have to take time to swap instructions in and out of RAM. It is usually 5 to 10 times faster than main memory. The speed of the CPU is faster than RAM. Hence, cache memory is placed between CPU and RAM so that the speed of operation of main memory and cache memory together can meet the speed requirements of the high-speed CPU.

Buffer Memory

Buffer is a part of memory which stores data temporarily while that data is being forwarded from one location to another in a computer. It is normally used between the CPU and the input/output devices. It acts as an intermediary between two processes or devices that may operate at different speeds or have different data transfer rates. Few examples where buffer memory are used: Video Streaming, in a printer spooler, operating system file operations

Secondary Memory

Secondary memory is the non-volatile memory which is used to store data and instructions permanently even when power is turned off. It is also called as auxiliary memory or back-up memory. Secondary memory is slower than primary memory. It is the memory that cannot be directly accessed by the computer CPU. Following are the types of secondary memory: -

- Magnetic Memory
- Optical Memory
- Flash Memory

Magnetic Memory

Magnetic memory is a type of secondary memory coated with magnetic sensitive material such as iron oxide at its surface.

1. **Hard Disk**

   Hard disk is a magnetic disk that is used as secondary memory for mass storage of data permanently. It stores programs, data, operating system, compilers, assembler, application program and database. It consists of spinning platters coated with a magnetic material and a read/write (R-W) head that moves to access data.

2. **Floppy Disk**

   Floppy disk (also known as diskette), is a plastic disk coated with magnetic material and covered by a protective jacket. It is portable and inexpensive. It comes with very less storage capacity i.e. 360 KB to 1.2 MB. It is mainly used for transferring data from one computer to another. It requires floppy disk for its operation. It is rarely used at present time.

3. **Magnetic Tape**

   Magnetic tape is a sequential access storage device made up of thin plastic ribbon/tape coated with magnetic material. It is mainly used for storing audio, video and keep backup. Magnetic tape drive used for reading or writing data.

4. **Magnetic Drum**

   Magnetic drum is a magnetic data storage device which was the early form of computer's memory. It was invented by Gustav Tauschek in 1932 in Austria. It was widely used in the 1950s-1960s as both primary and secondary storage in early computer system.

Optical Disk

1. **CD-ROM**

   CD-ROM stands for Compact Disk Read Only Memory. It is an optical storage device used to store text, graphics, sound and video. It can store about 700 MB of data or 80 minutes of video. It is very useful in storing large amount of data like computer software, movies, audio or video songs and so on. Other types of CD-ROM are CD-R (Compact Disk Recordable) and CD-RW (Compact Disk Re Writable).

2. **DVD**

   DVD stands for Digital Versatile Disk or Digital Video Disk. It is also a type of optical storage device which has a very large storage capacity. It is also considered as the improved form of CD-ROM. It can store about 4.7 to 17 GB of full-length movies, games and software. DVD-R, DVD-RW and DVD-RW are examples of DVD.

3. **Blu-Ray (BD)**

   BD is the name of the new generation optical disk. It is developed to enable recording, rewriting and playback of high-definition video (HD), as well as storing large amount of data. This disk offers more than five times the storage capacity of present DVDs. It can hold up to 25 GB on a single layered disk and 50 GB on a dual layered disk.

Flash Memory

Flash memory is a non-volatile memory storage medium made up of semiconductor chips. Flash memory is developed by Japanese engineer Dr. Masuoka Fujio and his team at Toshiba Corporation in the mid-1980. It stores and transfers data between digital devices. To transfer and store data, it needs to be connected to other compatible devices through their USB ports. They are commonly found in a range of portable devices, like USB flash drives, smartphones, digital cameras, video games, tablet computers, flash memory cards and SD cards. Flash memory is a type of **EEPROM** chip, which stands for **Electronically Erasable Programmable Read Only Memory**. It has a grid of columns and rows with a cell that has two transistors at each intersection

Difference between Primary and Secondary Memory.

Primary Memory	Secondary Memory
Primary memory is the main memory of the computer.	Secondary memory is the backup memory of the computer.
It is directly associated with CPU.	It is not directly associated with CPU.
It is faster than secondary memory.	It is slower than primary memory.
It can store small amount of data.	It can store large amount of data.
It is expensive as compared to secondary memory.	It is less expensive as compared to primary memory.
It stores data temporarily.	It stores data permanently.
For example: RAM, ROM etc.	For example: Hard disk, Floppy disk, CD, DVD

Input Devices

Input device is a hardware component that allow users to feed data or instructions into a computer system. They translate human actions or physical conditions into signals that a computer can understand and process.

Some of the commonly used input devices are keyboard, mouse, scanner, light pen, microphone, Joystick, Trackball, OCR, OMR, BCR, MICR, Touchpad, Touch Screen, Digital Camera etc.

1. **Keyboard**

   Keyboard is the most common input device used to input data into the computer. It looks like an electronic typewriter keyboard but contains keys for specific purposes. QWERTY keyboard is commonly used.
   Most keyboard have the following keys.
   Alphanumeric keys (Letters and Number)
   Punctuation keys (commas, Semicolon and so on)
   Special Keys (Function keys, Control keys, arrow keys, cap lock key and so on)

2. **Mouse**

   Mouse is a pointing input device which is used to control the position of the cursor on the screen. It contains two or three buttons on the top. By pressing the button of the mouse user can communicate to the computer
   Type of Mouse:
   Mechanical Mouse: Mechanical mouse had a rubber or metal ball that detects the movement and direction as it rolls.
   Optical Mouse: Optical mouse uses laser light to detect the movement.

3. **Trackball**

   Trackball is a pointing device similar to a mouse. It has a movable ball on top that rotated with the fingers or palm of the hand. Instead of moving the mouse around on the desk, the trackball is moved with the tips of the finger. Trackball is mostly used in portable computers like laptop.

4. **Joystick**

   Joystick is a pointing device used to control the movement of the cursor on the computer. It consists of a vertical handle like a gearshift lever mounted on a base with one or two buttons. It is mainly used while playing games to control the movement of characters or objects.

5. **Light Pen**

It is a light sensitive pen-like input device and is used to select objects directly on the computer screen. It is used for making drawing, graphics and for menu selection. Figures and drawings can be made by moving the pen on computer screen. It is useful for graphics work like CAD package and is mainly used by engineers, graphic designers and illustrators.

6. **Touchpad**

A touchpad is a pointing input device commonly used on laptops. It is used as an alternative of mouse. It allows the user to move a cursor with their finger. It can be used in place of mouse.

7. **Touch Screen**

A touch screen is a display screen which allows users to interact with a computer using their finger or stylus. It uses optical sensors to detect the touch of a finger on the screen. User can open different programs by touching specific points or program icons. It is very popular and used on variety of devices such as computer, laptop displays, smartphones, tablets etc.

8. **Digital Camera**

Digital camera is used to capture pictures and video which is in the digitized form and can be stored in computers. It can take photo and load them directly from the camera digitally to a main storage or secondary storage device.

9. **Scanner**

Scanner is an input device which is used to enter text, images, and graphs printed on paper directly into the computer. It electronically captures text, images and converted them into electronic format. Some of the popular scanners are Compact Desktop scanners and Flatbed scanners.

10. **Bar Code Reader (BCR)**

Bar code reader is a scanner used for reading or decoding barcode information. Bar code is printer series of parallel bars or line of varying width and spacing to indicate some specific information. It contains identification of an item such as price, quality, manufacturer etc. It is mostly used in supermarket, bookshop, library etc.

11. Optical Character Reader (OCR)

OCR reads printed characters directly, converts them into appropriate ASCII codes and stores them in a computer. It uses light and sensor to determine the character. OCR devices are used in telebanking, Airlines, Postal offices etc.

12. Optical Mark Reader (OMR)

OMR is a device that reads pencil marks or shades and converts them into computer usable form. It is mainly used to read answer sheets of multiple choice questions and survey form, Graduate Record Examination etc.

13. Magnetic Ink Character Reader (MICR)

MICR is an input device, which reads the character printed using magnetic ink. Magnetic ink is a special type of ink containing iron-oxide particles, which is used to write characters. MICR is mostly used in bank for processing cheque.

14. Digitizing Tablet

A digitizing tablet consists a tablet connected by a wire to a stylus. A stylus is a pen-like device with which the user sketches an image. It is mainly used in design and engineering.

15. Smart Card

A smart card looks like a credit card but contains a microprocessor and memory chip. It can be used for using public telephone, bank card, as well as for carrying their medical histories on them.

16. Webcam

Webcam is an input device capable of capturing digital images. It is a is a small video camera connected to a computer via USB, used for capturing and transmitting video over the internet. It's commonly used for video conferencing, live streaming, and real-time communication.

17. Microphone

Microphone is an input device used for recording voice or sound. It is widely used for the sound input to the computer for voice mail, videoconferencing, video chat etc.

Output Devices

An output device is a hardware device that is used to show the processed results to the user in the form of text, audio, video, visuals on a computer screen or a printed hard copy on paper. Some of the output devices are monitor, printer, plotter, projector, speaker etc.

Based on the output produced, output devices are categorized into following types:-

- Softcopy output devices
- Hardcopy output devices

Softcopy Output Devices

Softcopy refers to intangible output mainly displayed on the screen or through other output devices such as speakers. It is temporarily and cannot be touched.

Monitor

A monitor is an output device that displays video images and text. It is also known as Visual Display Unit (VDU). Monitors are similar to **TVs**, but they are designed to have a **higher resolution**, meaning they show much sharper and clearer images.

On the basis of working principle, it is categorized into three types: -

CRT (Cathode Ray Tube)

CRT monitors are the traditional technology used in older TVs and computer displays. It is bigger and bulky in size. They use a cathode ray tube to project images onto the screen by firing electron beams at phosphorescent material. The CRT monitor despite being bulky has some advantages such as high contrast ratio and color depth because of which it is still preferred by graphic designers.

LCD (Liquid Crystal Display)

LCD monitors use liquid crystals placed between two layers of glass or plastic, with a backlight to display images. It creates images with a special kind of liquid crystal that is normally transparent but becomes opaque when charged with electricity. LCD is lighter in weight, not bulky but expensive compared to CRT monitor. It consumes low voltage and power consumption. The user should sit in front to monitor, if he sits with angle, content is not visible.

LED (Light Emitting Diode)

An LED monitor is a screen that uses light-emitting diodes (LEDs) to produce images and videos. LED monitors are better because they use less energy and are thinner than older screens. They also give brighter and clearer pictures with 180˚. That's why many people use LED monitors for gaming, watching videos, and more.

Difference between CRT and LCD monitor.

CRT	LCD
CRT stands for Cathode Ray Tube.	LCD stands for Liquid Crystal Display.
It requires more desk space.	It requires less desk space.
It consumes more power.	It consumes less power.
It is heavier, bulky and large in size.	It is lighter, compact and thin in size.
It is less expensive.	It is more expensive.
It emits harmful radiation.	It doesn't emit radiation.
It is used only for PCs.	It is used for Laptop, tablets, smart phones.

Speaker

A speaker is an output device that is connected to the computer to generate sound. It converts electrical signals into sound waves. It allows user to hear audio output from various devices such as computers, smartphones, televisions, and music players. Sound card enables the computer to output sound to audio devices, as well as accept input from a microphone.

Hardcopy Output Devices

A **hardcopy** is a tangible output produced mostly on papers by devices such as printers. It is permanent and can be touched. The most common hardcopy output devices are printer and plotter.

Printer

Printer is the most popular and commonly used hardcopy output device. It prints the output on the paper. On the basis of working principle there are two types of printers they are: -

- Impact printer
- Non-Impact printer

Impact Printer

Impact printers are the printers which have working mechanism similar to a typewriter i.e. it forms characters or images by striking print hammer or wheel against inked ribbon. Print hammer is responsible for transferring the ink to the paper in the form of required characters. The example of impact printers are Dot matrix printer, Daisy Wheel printer, Line printer.

Dot Matrix Printer

It prints text and graphics on paper by physically striking pins against an inked ribbon. The inked ribbon is placed between the print head and the paper. Its printing speed is very slow and does not provide high quality printing. It produces noise while printing. Printing cost is cheaper as it uses the simple technology. It can print multiple copies of output at a time by using carbon paper. Hence it is very useful for printing bills.

Daisy Wheel Printer

Daisy wheel printers are the impact printers very popular during 1970s and 1980s. Daisy wheel is a disk made of plastic or metal containing letters, numbers and other characters. To print characters, printer rotates the disk until the desired letter is facing the paper then strike with a hammer. It is noisy and slow. It has a fixed font and cannot print graphics.

Non-Impact Printer

Non-Impact printers are faster and quieter than impact printer because they have fewer moving parts. It forms characters and images without making direct physical contact. They spray ink on the paper with the help of nozzle. The examples of non-impact printers are laser printer, ink jet printer, thermal printer.

Laser Printer

A laser printer is a type of non-impact high resolution printer that uses a laser beam to produce high-quality text and graphics on paper. Laser printer is more expensive in comparison to the other printer. The printing speed of this printer is high. Laser printers are used in offices, school, and other environments where quick and large volume printing is required.

Ink-Jet Printer

Ink-jet printer is a type of non-impact printer that prints text and graphics on the paper by spraying small drops of ink on a paper with the help of nozzle. It is faster less noisy and produces very high quality of print than impact printers but its printing cost is higher. It can print both monochrome and chrome. Inkjet printer is mostly used in the home and small offices usages.

Difference between Impact and Non-Impact printer.

Impact Printer	Non-Impact Printer
It prints characters or images by striking print hammer or wheel against inked ribbon.	It prints characters and images without striking the papers.
Its speed is slower.	Its speed is faster.
Its printing quality is lower.	Its printing quality is higher.
It normally uses continuous paper sheet.	It normally uses individual paper sheet.
It produces noise during printing.	It does not produce noise during printing.
It uses inked ribbon for printing.	It uses toner or cartridge for printing.
It is less expensive.	It is more expensive.
For example: Dot matrix printer, Daisy Wheel printer.	For example: Laser printer, Ink-jet printer.

Plotter

Plotter is also a hardcopy output device which is used to produce good quality graphics or drawing such as pie-chart, bar graph, map etc. on paper. It uses ink pen or ink jet to draw graphics or drawing. It is mostly used in engineering field.

Interfaces

A hardware interface is defined as the point where electronic components interact within a computer system, facilitating the exchange of information between these components. It is also called port.

Parallel Port

A parallel port is a type of computer interface that allows you to connect devices for data transfer. It was commonly found on older computers and used for connecting portable hard disk, scanner or CD-ROM and other

peripherals. There are multiple lines or wires to connect the peripherals to the port in parallel port.

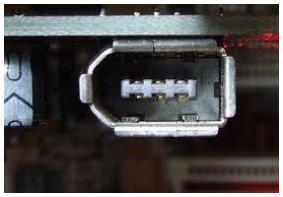
Serial Port

A serial port is an interface used for communication that sends data in a sequence, one bit at a time. It is one of the oldest types of interfaces and at one time was commonly used to connect printers. Serial ports are designed for long-distance communication and are commonly used with devices that don't require high-speed data transfer.

Universal Serial Bus (USB) Port

USB stands for 'universal serial bus'. It is the universal port and it is used to connect different peripherals to a computer system. Data and information flow in and out through the universal port. It is the high speed serial bus. Many peripherals such as USB keyboard, USB mouse, digital camera, web cam, can be connected to the USB port.

IEEE 1394 Interface Port



IEEE-1394 is a communication technology that was developed by apple in the early 1990s. The IEEE 1394 port is a serial bus interface standard for high speed communication for digital audio, digital video etc. It is also known as firewire. Because it was more costly to implement than USB, IEEE did not become as popular an interface for peripheral devices.

HDMI Port

High definition multimedia interface, HDMI is a connector and cable capable of transmitting high-quality and high-bandwidth streams of audio and video between devices. The HDMI technology is used with devices such as an HD TV, projector, DVD player, or Blu-ray player. They can run from one foot up to 50 feet, though it's not recommended that users buy more than a 25 foot cable as it may result in signal degradation or loss.